

WADENA, MN, USA

WMO: 726561

Lat: 46.447N

Lon: 95.212W

Elev: 418

StdP: 96.41

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 94997

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.0	(c) -27.2	(d) -32.5	(e) 0.2	(f) -29.2	(g) -29.9	(h) 0.2	(i) -26.4	(j) 11.2	(k) -11.3	(l) 10.2	(m) -11.4	(n) 2.2	(o) 260	(p) 0.620

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.9	(c) 30.0	(d) 22.0	(e) 27.9	(f) 21.0	(g) 27.2	(h) 20.2	(i) 23.8	(j) 28.2	(k) 22.5	(l) 26.7	(m) 21.2	(n) 25.4	(o) 5.0	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.4	18.0	26.7	21.2	16.7	25.4	19.8	15.3	23.9	73.2	28.3	68.4	26.4	63.7	24.5	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.0	(b) 8.7	(c) 7.9	(e) -33.5	(f) 32.8	(g) 2.9	(h) 1.7	(i) -35.6	(j) 34.0	(k) -37.3	(l) 35.0	(m) -38.9	(n) 36.0	(o) -40.9	(p) 37.2		
(4) 10.0	(5) 8.7	(6) 7.9	(7) -33.7	(8) 25.3	(9) 2.8	(10) 1.4	(11) -35.7	(12) 26.3	(13) -37.4	(14) 27.1	(15) -39.0	(16) 27.9	(17) -41.0	(18) 28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-13.0	-11.8	-3.4	4.7	11.6	17.1	19.8	18.2	14.2	6.1	-2.1	-9.9		
	DBStd	12.91	7.87	7.09	7.43	5.80	4.92	3.53	3.16	3.32	4.87	5.11	6.04	6.93		
	HDD10.0	3095	714	612	417	174	39	1	0	1	16	143	363	616		
	HDD18.3	5252	973	845	673	410	217	63	21	42	142	379	612	874		
	CDD10.0	1041	0	0	2	14	87	213	306	256	140	23	1	0		
	CDD18.3	156	0	0	0	0	7	25	68	39	16	1	0	0		
	CDH23.3	1558	0	0	0	9	98	244	637	387	169	13	0	0		
	CDH26.7	367	0	0	0	2	23	47	162	93	38	2	0	0		
(14) Wind	WSAvg	3.4	3.4	3.4	3.6	4.0	3.9	3.2	2.7	2.6	3.2	3.5	3.5	3.4		
(15) Precipitation	PrecAvg	705	20	18	35	60	83	113	100	77	73	75	32	19		
	PrecMax	862	70	51	91	125	180	185	212	189	170	176	101	43		
	PrecMin	553	2	4	12	25	18	44	33	17	26	8	1	7		
	PrecStd	96	16	12	18	27	39	41	42	48	34	49	25	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	7.2	17.8	23.0	29.1	31.0	32.7	32.1	30.9	24.9	16.3	6.3		
		MCWB	1.8	5.0	12.3	14.6	18.9	22.9	24.3	23.6	21.9	16.2	10.4	4.0		
	2%	DB	1.3	2.9	13.6	19.8	26.3	27.7	30.0	28.8	27.5	21.1	12.3	2.6		
		MCWB	0.1	1.3	9.2	12.0	17.2	20.6	22.8	21.8	20.3	14.6	8.2	1.1		
	5%	DB	-0.9	1.1	9.2	17.3	22.9	26.2	27.9	27.4	25.2	17.4	8.7	1.0		
		MCWB	-1.3	0.1	6.5	10.0	15.3	18.8	21.6	20.6	18.6	12.4	5.9	0.1		
	10%	DB	-2.3	-1.3	6.3	14.1	21.0	24.0	27.2	25.9	22.6	13.9	6.3	-1.0		
		MCWB	-2.9	-1.9	3.5	8.1	14.4	17.9	20.8	19.4	17.7	9.9	3.7	-1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	5.1	13.5	15.3	20.6	24.5	25.8	24.6	23.6	17.8	11.6	4.0		
		MCDB	4.3	7.0	16.4	23.9	27.0	28.7	30.1	29.4	27.9	22.5	15.0	5.5		
	2%	WB	0.2	1.7	9.7	12.5	18.4	22.2	24.2	23.0	21.5	15.2	8.4	1.4		
		MCDB	1.3	3.1	12.4	18.0	23.5	25.9	28.6	27.1	25.6	20.0	11.2	2.5		
	5%	WB	-1.2	-0.1	6.9	10.6	16.8	20.5	22.9	21.7	19.9	12.6	6.0	-0.1		
		MCDB	-0.7	1.0	9.9	16.0	21.5	24.8	27.2	25.9	23.9	16.1	8.5	0.6		
	10%	WB	-2.7	-2.3	3.8	8.9	15.0	19.0	21.5	20.3	18.5	10.7	4.0	-1.3		
		MCDB	-2.4	-1.8	6.1	14.5	19.6	23.0	25.5	24.3	21.6	13.6	6.2	-0.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.5	10.2	11.4	11.9	11.2	11.9	12.1	11.9	9.9	8.7	8.4		
		MCDBR	9.5	10.6	13.5	16.4	16.0	13.5	13.4	13.9	14.8	15.0	13.2	9.9		
		MCWBR	8.2	8.8	9.4	10.0	8.8	7.3	7.1	7.3	7.9	9.2	9.0	8.3		
		MCDBR	9.1	9.8	13.2	14.9	13.8	11.5	12.3	12.1	12.4	13.4	11.9	9.1		
	5% WB	MCWBR	7.9	8.2	9.5	9.4	8.3	7.2	7.3	7.0	7.4	8.7	8.6	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.263	0.298	0.302	0.356	0.375	0.378	0.381	0.383	0.352	0.309	0.279	0.258		
	taud		2.362	2.257	2.395	2.328	2.338	2.362	2.371	2.375	2.443	2.544	2.516	2.412		
	Ebn at Noon		845	871	928	899	890	887	879	862	863	861	827	813		
	Edh at Noon		71	98	100	118	121	119	117	112	96	74	62	61		
(44) All-Sky Solar Radiation	RadAvg		1.61	2.73	3.83	4.84	5.37	5.89	6.26	5.27	3.94	2.44	1.51	1.11		
	RadStd		0.19	0.29	0.38	0.40	0.39	0.39	0.27	0.46	0.30	0.38	0.15	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page